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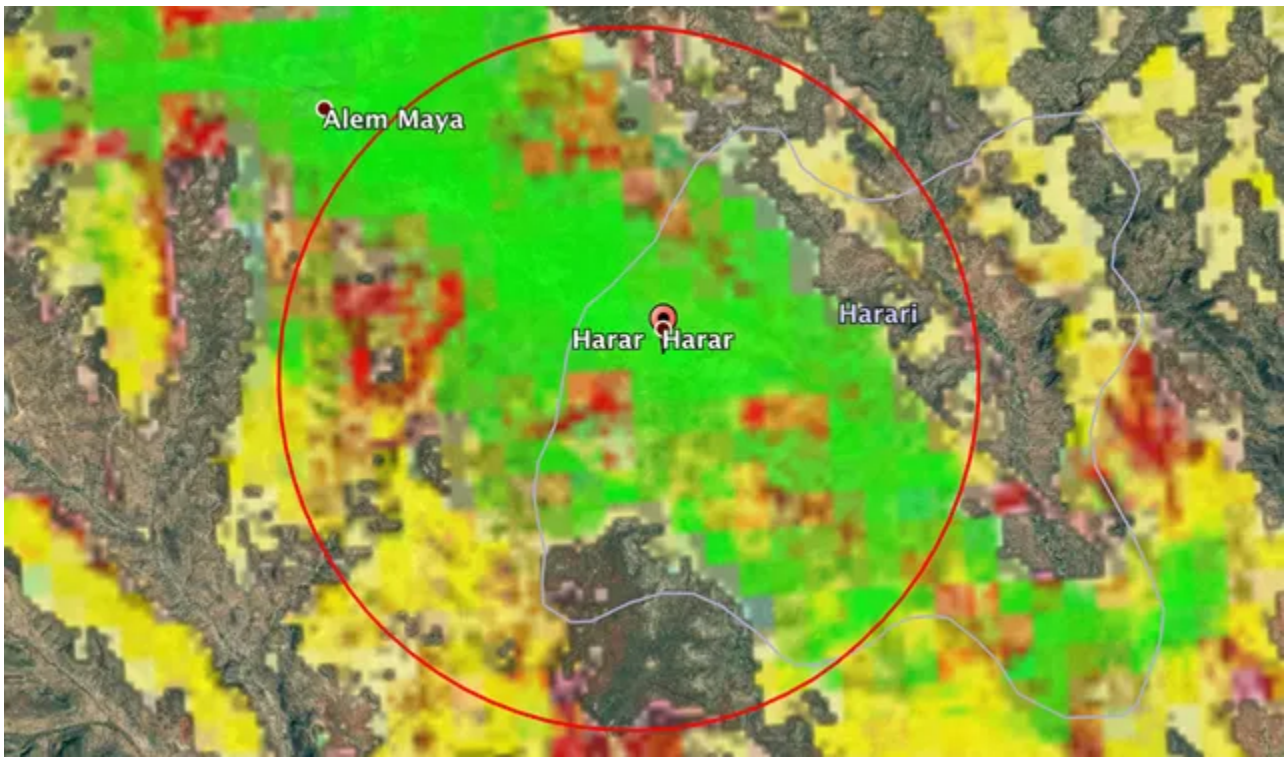
Finally! Accurate Mobile Network Coverage Maps Show Connectivity Reality

By Troy Etulain on January 7, 2019



Closing the global digital divide is such a daunting challenge, some companies have resorted to

expensive, eyebrow-raising, flying moonshot solutions. Despite the boldness of these attempts, the most exciting advancements increasing mobile and internet access have actually been small, cost-effective base stations that usually don't get any higher than 10 meters.

The game changer has been the emergence of inexpensive and reliable equipment designed to serve smaller, lower-income communities national operators tend to leave out of their business models. Some examples of low-cost equipment include:

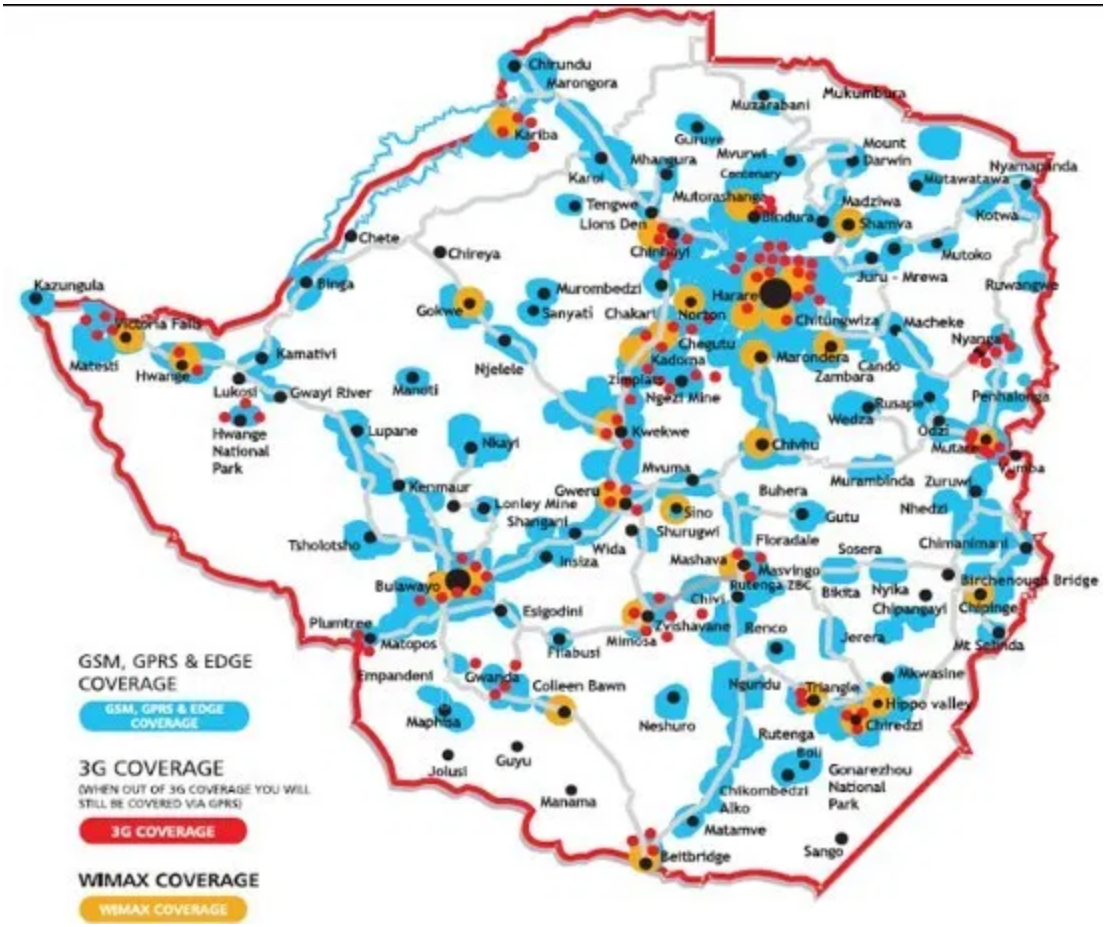
- Fairwaves UmSITE-TM3
- Nuran GSM LiteCell 1.5
- OpenCellular OC2G
- Sysmocom sysmoBTS 1002
- Vanu Community Connect

Where Are the Mobile Network Coverage Gaps?

While the software and hardware for these small base stations has been field-tested and is readily available, identifying ideal locations for its smaller coverage radius is often difficult, due to the lack of accurate coverage maps. We are in the process of choosing locations in northeast Nigeria for some equipment we recently won from the Telecom Infra Project (TIP), and have experienced the site selection challenge firsthand.

To date, most coverage maps that could provide accurate data have offered only a limited approximation of where coverage is available and have not accounted for population dispersion. Publicly available coverage maps often present mobile signal coverage as perfect circles based on tower location, despite the role trees or terrain have in blocking or weakening a cell tower's reach.

These generalized maps paint an overly optimistic picture of population centers with coverage. This often means both that people get left out and that it is hard to convince potential investors of actual opportunities in serving unconnected, albeit smaller populations. The true market size has been unappealingly opaque. Here is an example of a Zimbabwe map with circles over-stating coverage:



The obfuscation is so rampant and misleading that the FCC is now investigating coverage maps in the USA for potentially misleading the government about the reach of their wireless networks.

Mapping Mobile Network Coverage Gaps

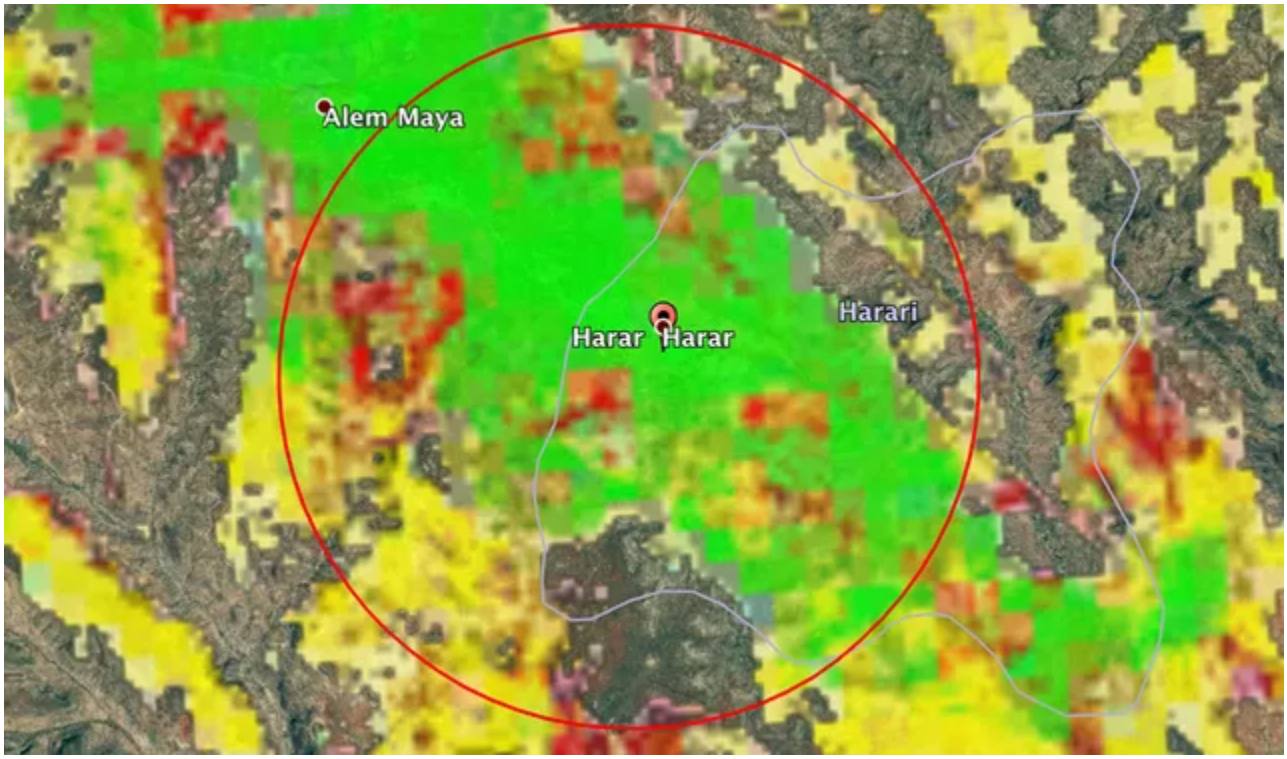
This traditional deficiency of coverage mapping is why FHI 360 was so excited to collaborate with Vanu Inc. to support the creation of unprecedentedly accurate coverage maps for all of Africa and many countries in Asia. The maps, which Vanu developed through advanced combination of open-source tools and data, enable us now to clearly see villages that could be profitably connected by small, cost-effective equipment.

While individual maps of similar accuracy have been produced for individual country markets, no one has ever created a single resource for an entire continent. And Vanu not only completed maps for the countries which development organizations care about in Asia, they can create a map for any new country with relative ease.

The Vanu maps easily show their usefulness in identifying people who currently lack connectivity and yet represent potential revenue for a carrier. Take for example the town of Harar (Harer) in eastern Ethiopia (the national market that lately has been garnering the most excitement).

One publicly-available coverage map shows the town covered by a perfectly round circle, despite the

hilly terrain. Vanu's, topographically-accurate map shows a strong signal in green, medium-strength signal in yellow, and a weak signal in red. It also shows there are actually multiple towers around Harar, and yet parts of the area lack any signal whatsoever.



Vanu estimates the perfectly round coverage circle over-estimates coverage by an astounding 90,000 people. It's striking to ponder how it would take for these people to be offered coverage if the operator thought they didn't need it.

Business Opportunities in Network Coverage Gaps

The level of mapping precision achieved by Vanu shows exactly which part of Harar lacks a signal—the population that would get left out if the existing coverage map were taken at face value. Layered with a population map, carriers, entrepreneurs and investors can accurately size the opportunity. And, as with any investment, the better the data, the greater the readiness to pursue an opportunity.

Inaccurate overstatements contribute to market failure. Using low-cost equipment from companies such as the ones listed above, it now only takes approximately 20% of the population in a village, comprised of even just 500 people, with $2 per month available to spend on communications to make a small site profitable.

When you consider that, according to the GSMA, unique SIM subscribers at the end of 2017 constituted only 44% of Africa's 1.2 billion population and, of this population, 63% lives in sparsely populated rural areas, you can begin to see how precise connectivity maps have immense potential for contributing to increased connectivity and revenue for those willing to provide it.

Around Africa, the rural connectivity opportunity captured in places like Harar is often pursued via deployment of what is called a wholesale business model. This works by one company installing and operating a network of mobile towers that route the signals of mobile network operators (MNOs). The MNO and the wholesale provider then share the revenue of traffic routed via the wholesaler's towers. Vanu, a hardware manufacturer and integrator of last-mile solutions, uses the wholesale operator model in multiple markets.

The wholesale model in is one of several we highlighted in our report Business Models for the Last Billion. For a long time, we, as well as others seeking to address the digital divide, have been working on various approaches to mapping connectivity. Better data on mobile coverage is particularly important for understanding the investment opportunity in areas with significant populations still seeking voice and broadband services. Coverage data enables precise calculation of the business opportunity available to connectivity entrepreneurs and investors.

To date, there are not many reliable and updated coverage maps available that present mobile and broadband coverage at the granularity required to understand the potential of network deployments that serve mostly rural populations (see, for example, the limited list here). The new connectivity maps developed by Vanu make a significant contribution to our available resources for granular coverage data. No single resource with this accuracy and scope has ever been put together.

Other Approaches to Producing Accurate Network Coverage Maps

Some other innovative approaches to coverage maps have been pursued or proposed, including:

- In Mapping the Unserved, Steve Song, with support from our USAID-funded mSTAR project, examined approaches to aiming to precisely identifying and quantifying populations without signal to build a picture of various countries' addressable markets.
- Recently, the GSMA Mobile for Development team began an effort, supported by DFID and USAID, to create connectivity maps using anonymized operator data and population data. So far, the GSMA has made Tanzania, Ghana and Nigeria available, with plans to release additional countries soon.
- A unique idea, originally proposed by Marco Zennaro of the International Centre for Theoretical Physics (ICTP), is to attempt to identify cell towers with high-resolution satellite imagery and the image recognition capabilities of machine learning, and then to build maps from there.

There are multiple ways of tackling the mapping challenge, and while the technologies in question will continue to evolve, we couldn't be more excited by Vanu's achievement. The maps' accuracy contribute to a message we try to convey whenever possible, that it is already possible to connect the unconnected, and to do so in the most sustainable way possible: *profitably*.

By Troy Etulain, Digital Development Department, FHI 360

Filed Under: Connectivity

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Written by Troy Etulain

Troy Etulain works as a Technical Advisor in FHI 360's Digital Development Department. Previously, Troy worked with UNHCR and USAID. He is passionate about connectivity, digital identities, and data for development. Opinions expressed here are his own and do not reflect the position of FHI 360 or other ICTWorks sponsors.

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5 Comments to "Finally! Accurate Mobile Network Coverage Maps Show Connectivity Reality"

1. *Charlene* says:
   January 7, 2019 at 5:38 am

   Great! but where do we find those VANU-FHI 360 maps of Africa? Are they not available for the public?

2. *Steve Hellen* says:
   January 7, 2019 at 11:43 am

   The original link given to the GSMA connectivity maps did not work; these maps are at https://www.mobilecoveragemaps.com/

   - *Charlene* says:
     January 8, 2019 at 3:40 am

     Awesome, thanks a lot. Not many countries in Africa yet but hopefully more coming soon!

3. *Sam Sheka Moi* says:
   January 7, 2019 at 9:41 pm

   Great

4. *Evert Bopp* says:
   January 16, 2019 at 5:51 am

Interesting development and great to see Open Source software being used for this type of work again.
We've done similar bur smaller scale work in post disaster response mapping cellular and wifi services and used Open Source software each time. Both because of cost but also because of functionality.
There's a blogpost from a few years ago giving a bit kore insight: http://disastertechlab.org/2014/05/23/wireless-site-surveys-in-disaster-zones/